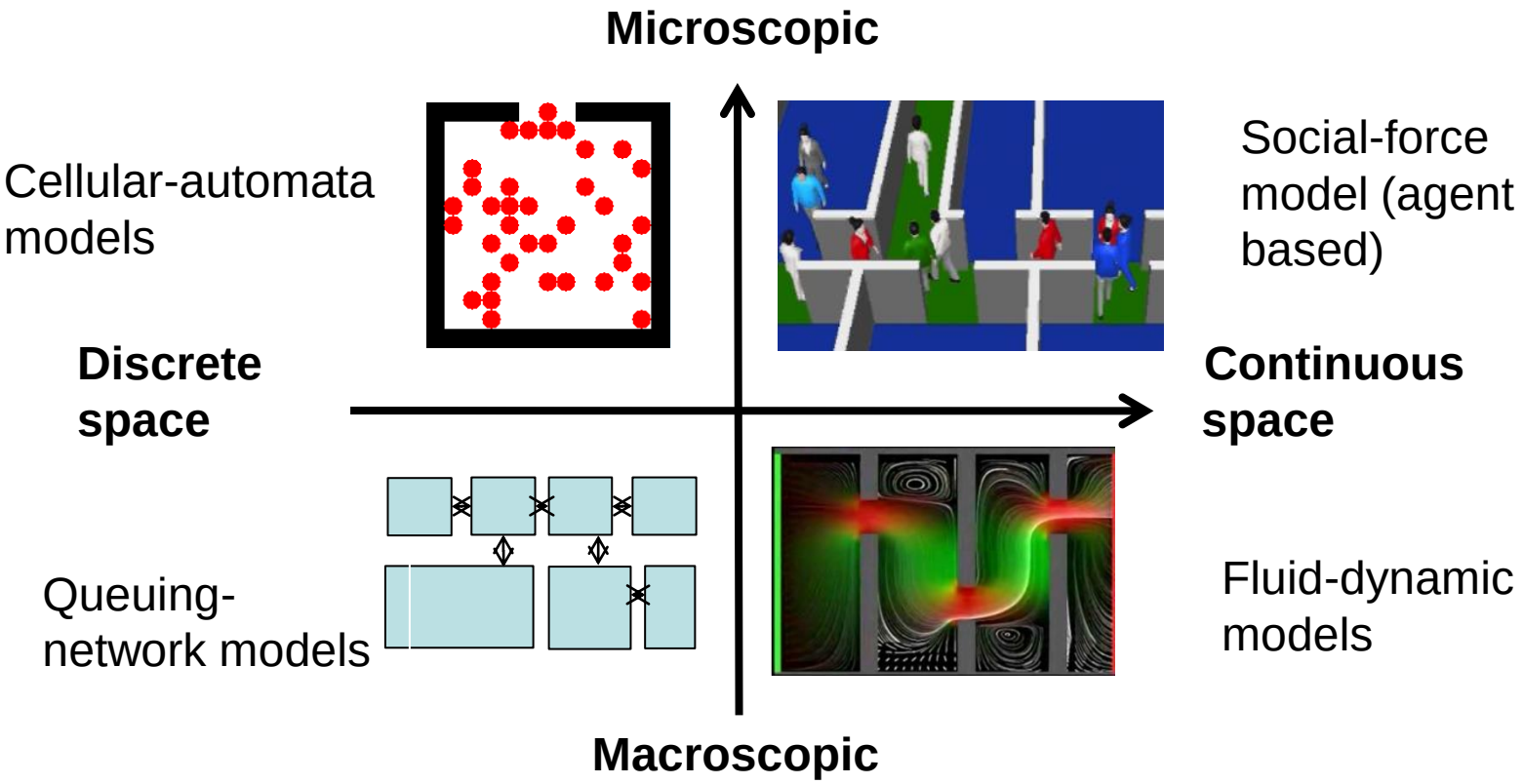
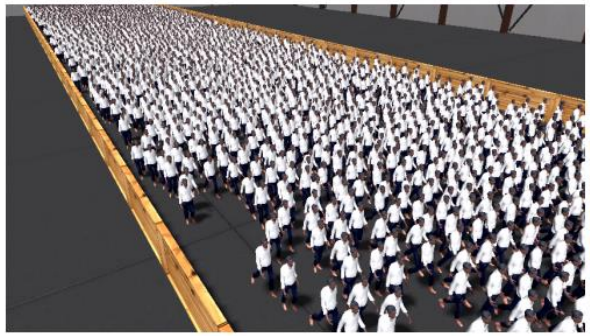
Introduction to crowd modelling

Simulating pedestrians can be done in various ways. Here is a rough classification of different simulation approaches.

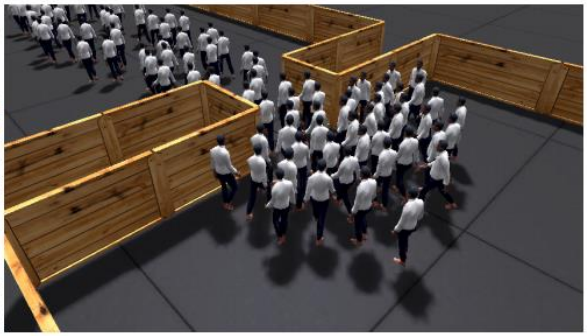


PLEdestrians

PLEdestrians, a least effort approach to crowd simulation (Guy et al., 2010)



(a) *Long Corridor*



(b) *Narrow Passage*



(a) *Simulated of Shibuya Station in Tokyo*



(b) *A different view of the simulated crossing*

Energy Minimization along a Line: We represent the line using the y-intercept form: $y = mx + e$. The velocity along this line that minimizes eqn. 4 can be computed using the following formulation. Let $\mathbf{v}_A^{new}$ be defined as an offset from a vector towards the goal:

$$\mathbf{v}_A^{new} = (\mathbf{G}_A - \mathbf{p}_A)/\tau + \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix}. \quad (6)$$

Let $(\mathbf{G}_A - \mathbf{p}_A)/\tau = (d_x, d_y)$. The magnitude r can be computed by solving the following quartic equation.

$$r^4 + Ar^3 + Br^2 + Cr + D = 0 \quad (7)$$

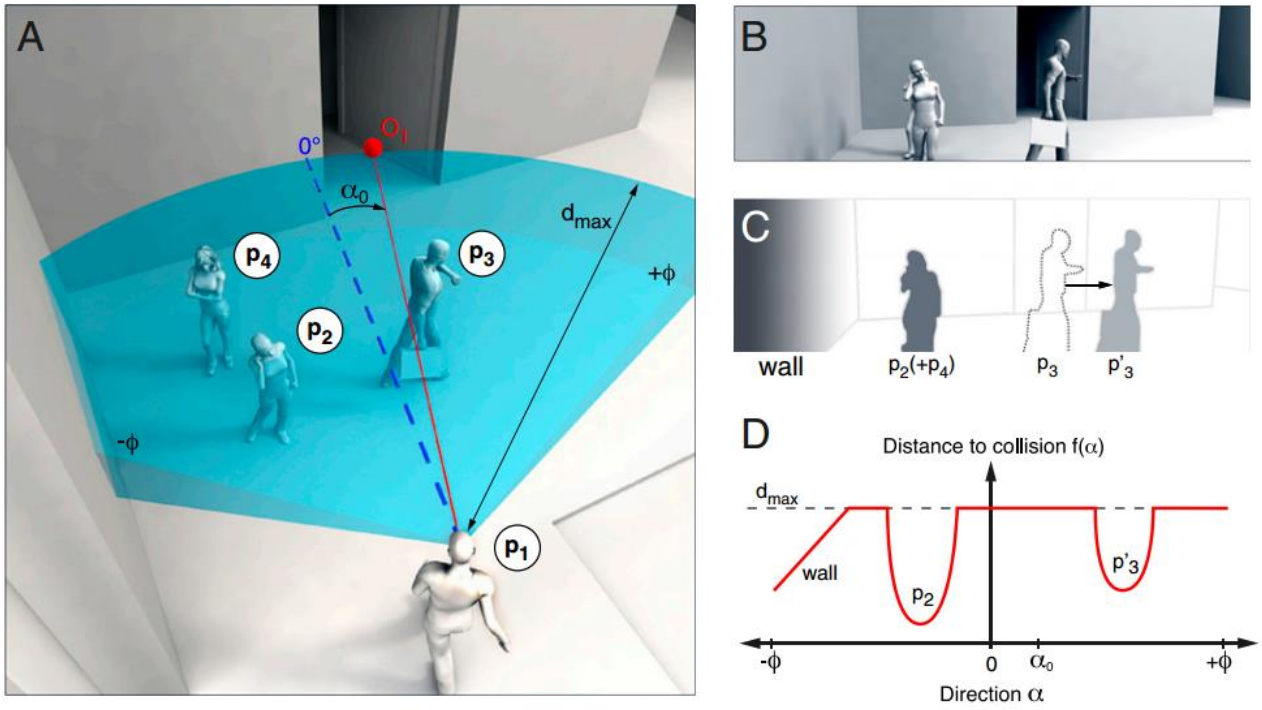
where:

$$A = \sqrt{\frac{4e_s}{e_w}}, \quad B = \frac{e_s}{e_w} + \frac{\tau^{-2}(d_x + md_y)^2(d_y - md_x)^2 - e^2}{(1+m^2)},$$

$$C = \frac{-2\sqrt{e_s}(\frac{d_y}{\tau} - m\frac{d_x}{\tau} - e)^2}{\sqrt{e_w}(1+m^2)}, \quad D = \frac{e_s(\frac{d_y}{\tau} - m\frac{d_x}{\tau} - e)^2}{e_w(1+m^2)}$$

Heuristics/vision based

Heuristics based crowd modelling (Moussaid et al., PNAS, 2011).

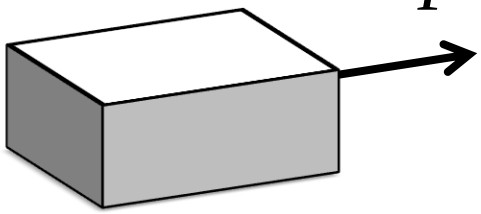


Social force(s) model

Sir Isaac Newton (1687):

The net force on an object is equal to the mass of the object multiplied by its acceleration.

$$\vec{F} = m\frac{d\vec{v}}{dt}$$



Kurt Lewin (1952): Behavioural changes are driven by social fields, or *social forces*.



Dirk Helbing (1990): Pedestrians are driven by social forces.

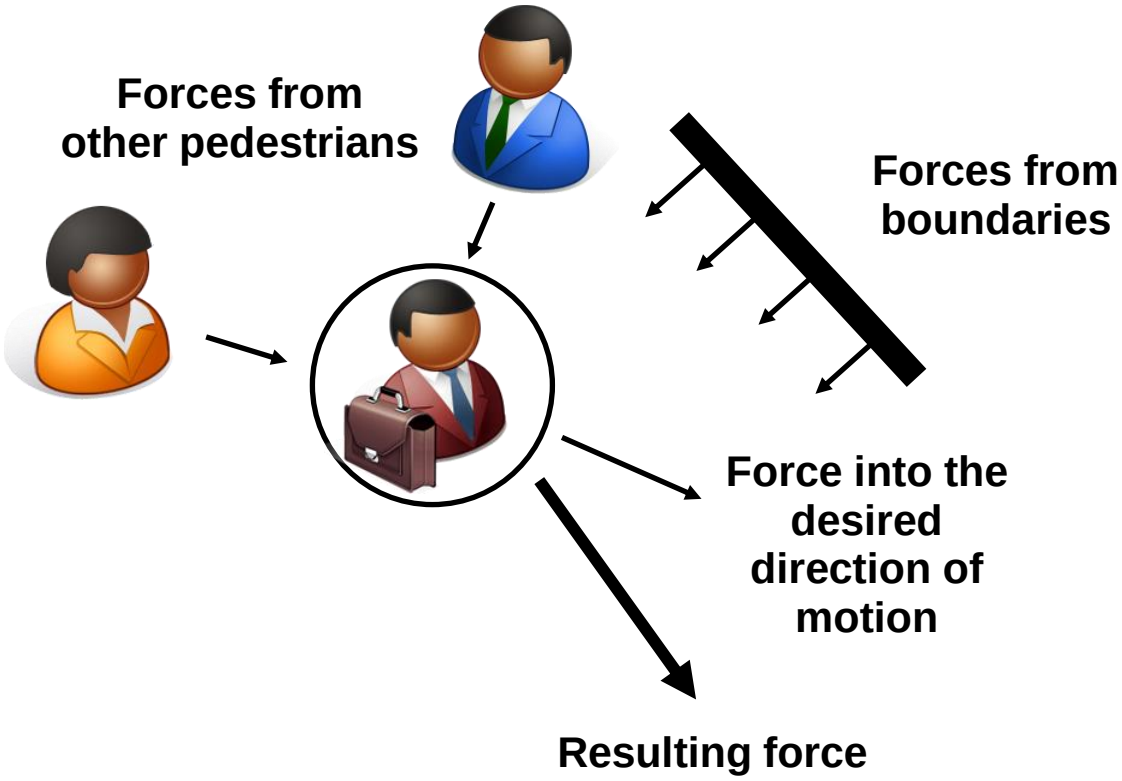
Social force model

The social-force model is a **microscopic model** which is **continuous both in space and time**. These properties makes the model suitable for reproducing dynamics realistically on a microscopic scale.



Social force model

Each pedestrian is influenced by a number of forces, e.g. repulsive forces from other pedestrians, repulsive forces from borders and a driving force towards the desired direction of motion.



Social force model

The Social-force model is specified via the equation of motion:

$$\frac{d\vec{v}_\alpha(t)}{dt} = \vec{f}_\alpha(t) + \vec{\xi}_\alpha(t)$$

Noise term

Where the force is composed by:

$$\vec{f}_\alpha(t) = \frac{1}{\tau_\alpha}(v_\alpha^0 \vec{e}_\alpha - \vec{v}_\alpha) + \sum_{\beta(\neq\alpha)} \vec{f}_{\alpha\beta}(t) + \sum_i \vec{f}_{\alpha i}(t)$$

Acceleration time

Desired velocity

Actual velocity

Forces from all other pedestrians β

Forces from all boundaries i

Social force model

Let us start with only the first force component, i.e. the adaptation of the *current* velocity to a *desired* velocity.

Varying parameter v^0 gives different desired speeds.

$$\vec{f}_\alpha(t) = \frac{1}{\tau_\alpha}(v_\alpha^0 \vec{e}_\alpha - \vec{v}_\alpha)$$

Acceleration time

Desired velocity

Actual velocity

Social force model

Varying desired speed v^0 :



Social force model

Let us start with only the first force component, i.e. the adaptation of the current velocity to a desired velocity.

If we assume that the desired velocity is known, the remaining parameter is τ.

$$\vec{f}_\alpha(t) = \frac{1}{\tau_\alpha}(v_\alpha^0 \vec{e}_\alpha - \vec{v}_\alpha)$$

Acceleration time

Desired velocity

Actual velocity

Social force model

Varying acceleration time τ :



Social force model

Next, we look at the second component, which reflects the forces from other pedestrians β.

$$\vec{f}_\alpha(t) = \frac{1}{\tau_\alpha}(v_\alpha^0 \vec{e}_\alpha - \vec{v}_\alpha) + \sum_{\beta(\neq\alpha)} \vec{f}_{\alpha\beta}(t)$$

Forces from all other pedestrians β

Social force model

If there is no interaction between pedestrians, we specify this component as:

$$\vec{f}_{\alpha\beta}(t) = 0$$

Social force model



Social force model

To apply the avoidance manoeuvre before the pedestrians collide, we add two terms; one physical force and one weaker (but longer range) social force:

$$\vec{f}_{\alpha\beta}(t) = A_p \frac{\vec{d}}{\left\|\vec{d}\right\|} \exp(-\left\|\vec{d}\right\| / B_p) + A_s \frac{\vec{d}}{\left\|\vec{d}\right\|} \exp(-\left\|\vec{d}\right\| / B_s)$$

The parameters are:

Physical: A_p = 2.0: interaction strength (m/s^2)
B_p = 0.1: interaction range (m)

Social: A_s = 0.5: interaction strength (m/s^2)
B_s = 3.0: interaction range (m)

Social force model



Social force model

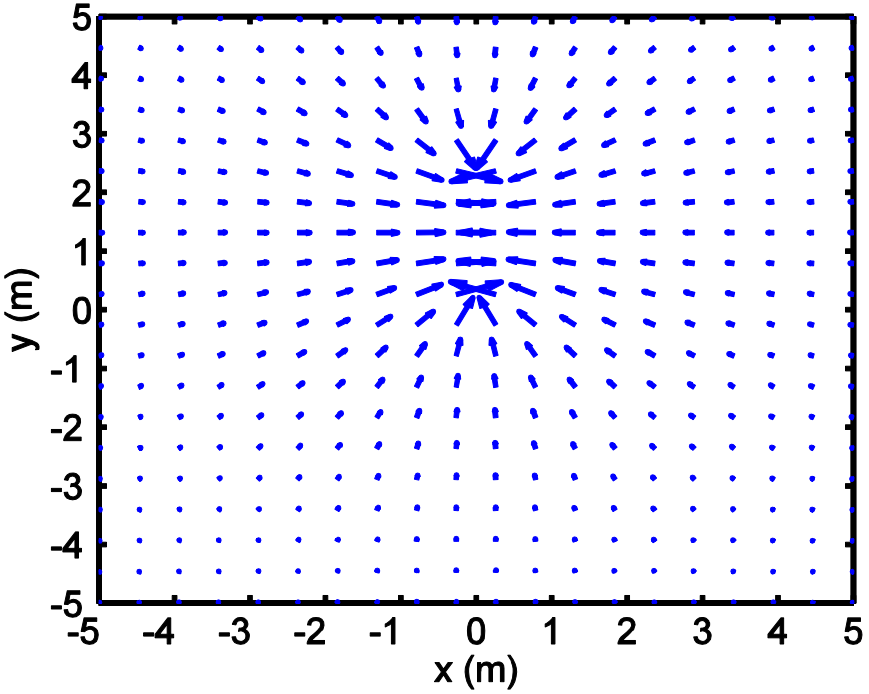


Pair-wide interaction: Improved specification

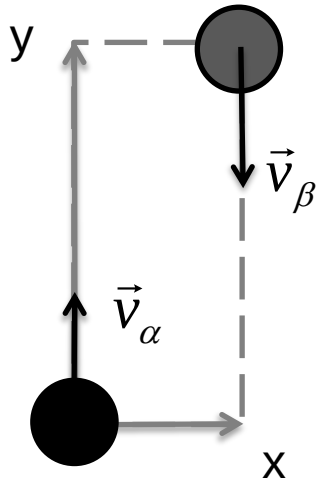
Improved, Elliptical Social-Force Model:

$$V_{\alpha\beta}(b) = AB\,\mathrm{e}^{-b_{\alpha\beta}/B}$$

$$2b := \sqrt{(\|\vec{d}_{\alpha\beta}\| + \|\vec{d}_{\alpha\beta} - (\vec{v}_\beta - \vec{v}_\alpha)\Delta t\|)^2 - [(\vec{v}_\beta - \vec{v}_\alpha)\Delta t]^2}$$



Fellow pedestrian β



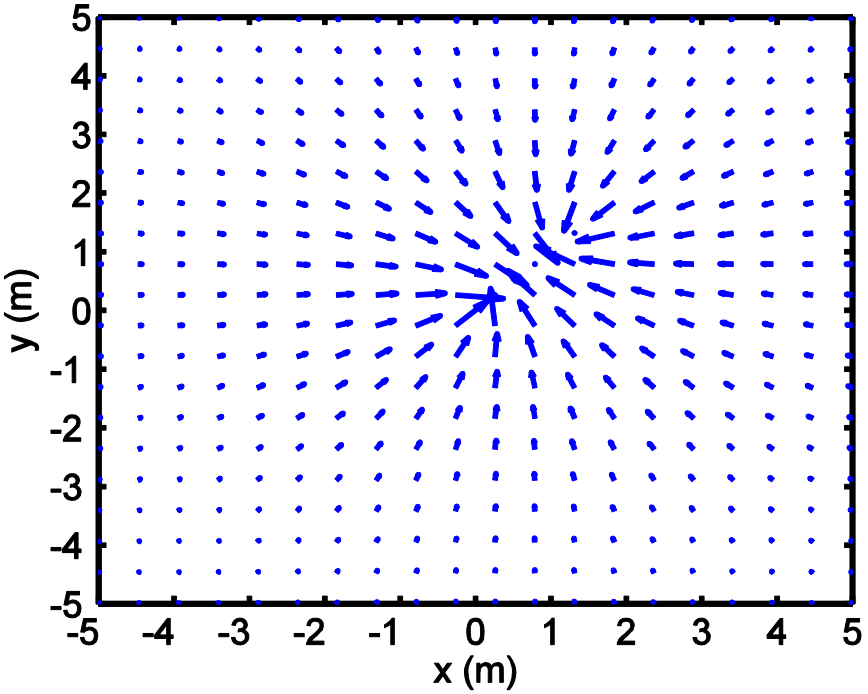
Pedestrian α

Pair-wide interaction: Improved specification

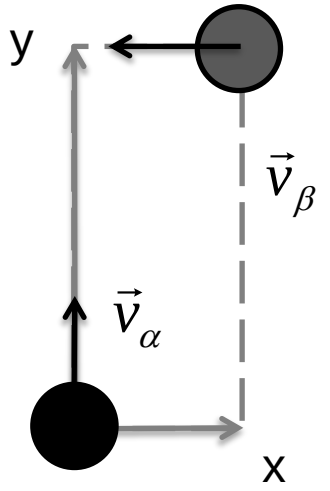
Improved, Elliptical Social-Force Model:

$$V_{\alpha\beta}(b) = AB\,\mathrm{e}^{-b_{\alpha\beta}/B}$$

$$2b := \sqrt{(\|\vec{d}_{\alpha\beta}\| + \|\vec{d}_{\alpha\beta} - (\vec{v}_\beta - \vec{v}_\alpha)\Delta t\|)^2 - [(\vec{v}_\beta - \vec{v}_\alpha)\Delta t]^2}$$



Fellow pedestrian β



Pedestrian α

Elliptical social force model



Social force model

Finally, forces from boundaries $\vec{f}_{\alpha i}(t)$ are needed to prevent the pedestrians from penetrating walls and obstacles.

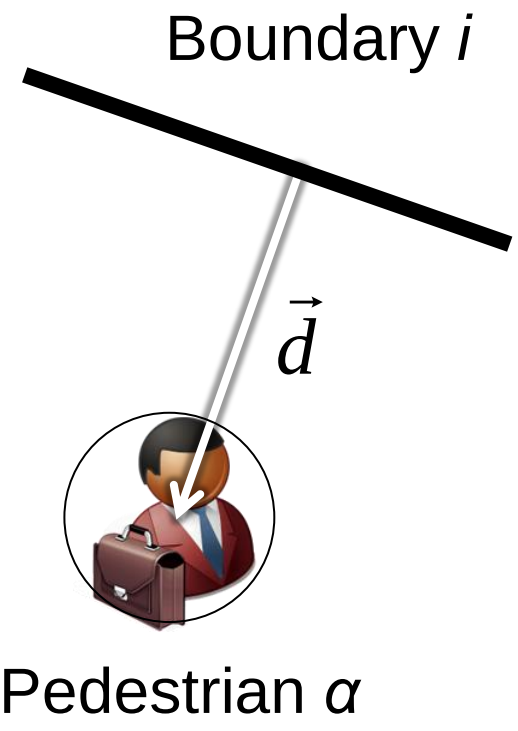
$$\vec{f}_\alpha(t) = \frac{1}{\tau_\alpha}(v_\alpha^0 \vec{e}_\alpha - \vec{v}_\alpha) + \sum_{\beta(\neq\alpha)} \vec{f}_{\alpha\beta}(t) + \underbrace{\sum_i \vec{f}_{\alpha i}(t)}$$

Forces from all boundaries *i*

Social force model

To prevent pedestrians from penetrating boundaries, we specify the force on the form:

$$\vec{f}_{\alpha i}(t) = A\frac{\vec{d}}{\left\|\vec{d}\right\|}\exp(-\left\|\vec{d}\right\| / B)$$



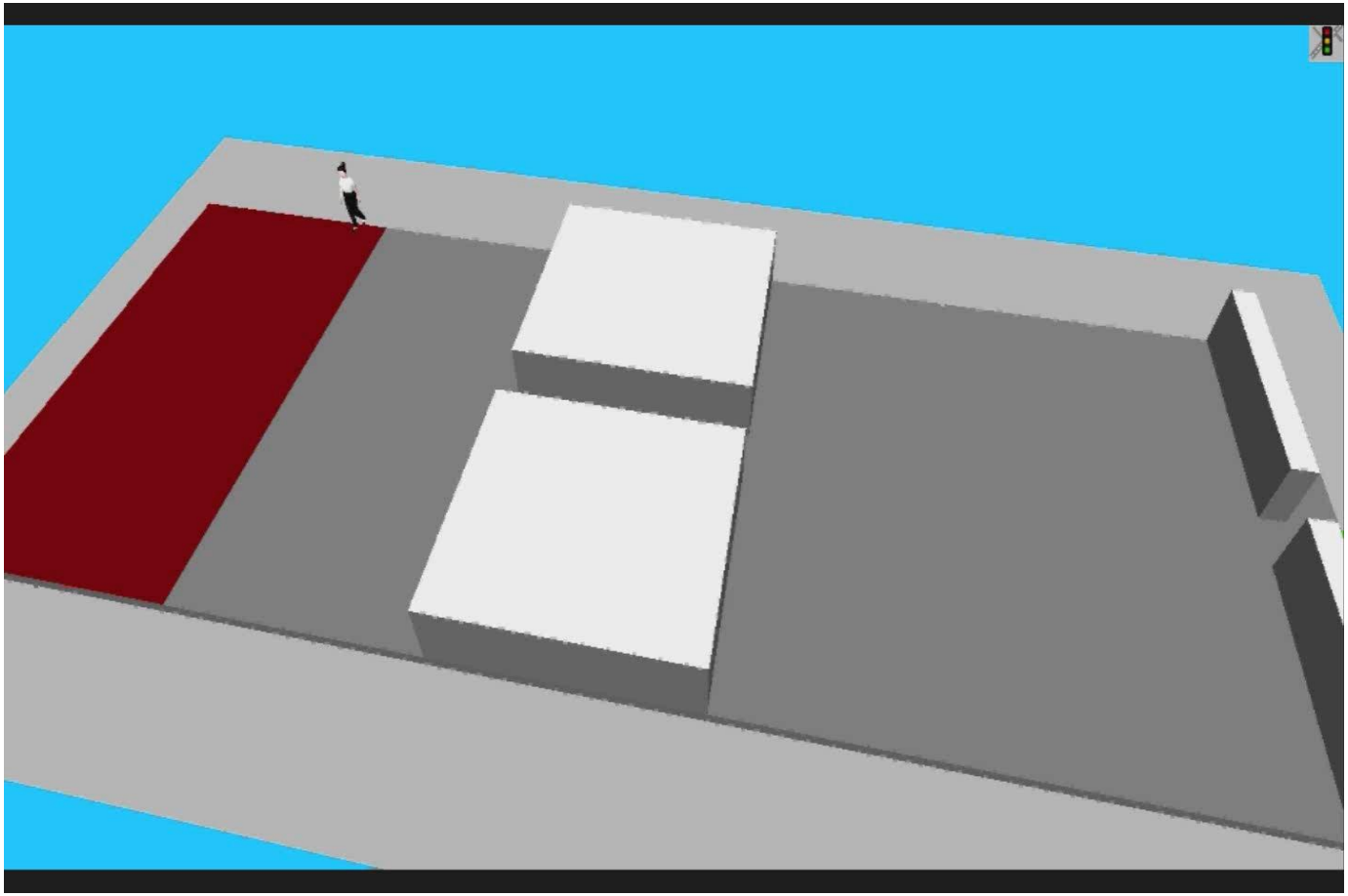
where $\vec{d}$ is the vector pointing from boundary i to pedestrian α.

There are two free parameters:

A=2.0: interaction strength (m/s^2)

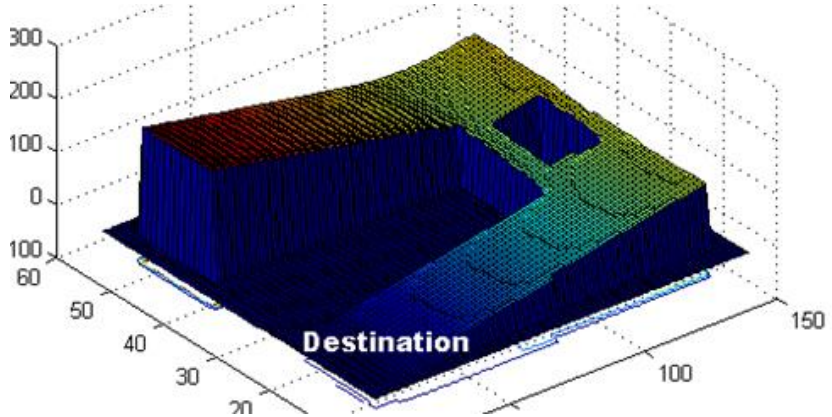
B=0.1: interaction range (m)

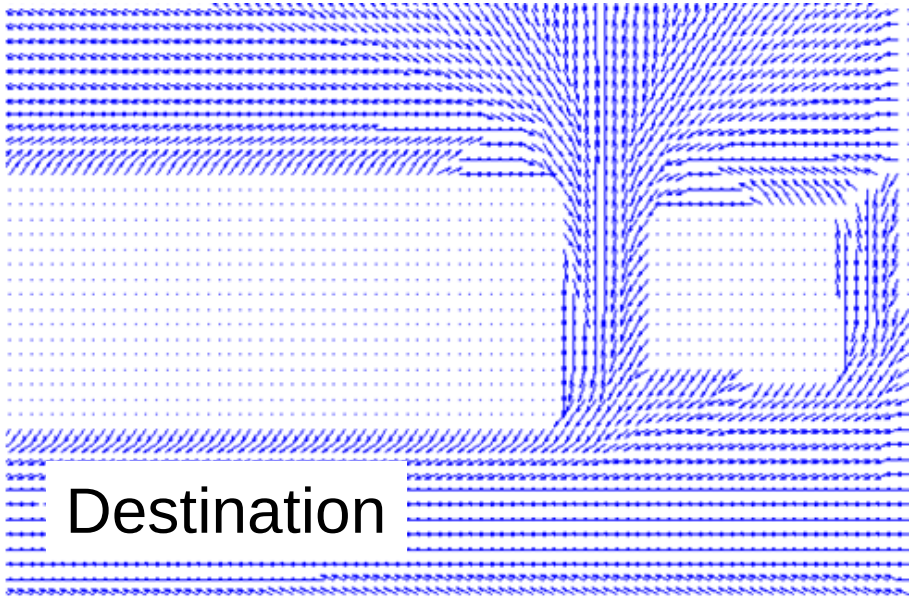
With boundary forces



Navigation

One way to achieve better navigation is to compute the distance to the destination from all locations, which can be done via the potential field. Technically, this is done by spreading a "wave" from the destination, similar to the "flood fill" method in painting applications.





Social-Force Model - Complex Environments

